DATASHEET

AprilGrid Calibration Target, Aluminum Composite

AprilGrid uses AprilTag 36h11 markers arranged in a regular grid. It is the standard target for Kalibr camera–IMU calibration and is also supported by the OpenCV ArUco module (`cv2.aruco.DICT_APRILTAG_36h11`). Each tag carries a unique ID, so partially visible boards still yield usable detections — useful at field edges where distortion is greatest.

1 Ordering

Part number	Size (W × H)	Tags (rows × cols)	Tag / gap	Markers
CT-APR-P-200X150-T36	200 × 150 mm	4 × 6	25 / 7.5 mm	24
CT-APR-P-300X200-T36	300 × 200 mm	5 × 7	30 / 9 mm	35
CT-APR-P-400X300-T36	400 × 300 mm	6 × 8	35 / 10.5 mm	48
CT-APR-P-600X400-T36	600 × 400 mm	7 × 10	45 / 13.5 mm	70
CT-APR-P-800X600-T36	800 × 600 mm	9 × 12	50 / 15 mm	108

Dictionary: AprilTag 36h11. Tag spacing ratio 0.3 (gap = 0.3 × tag side). IDs start at 0 at the bottom-left tag and run left to right, then upward, as Kalibr numbers them; the origin is the bottom-left tag corner, x right, y up. Kalibr `tagSpacing` parameter equals the gap divided by the tag side (0.3). Other grid sizes or spacing ratios: custom order (CUS).

Pattern area is centered with a white quiet zone ≥ 1 tag pitch on all sides; the edge strip with part number, serial and 100 mm scale bar sits outside the quiet zone.

2 Key specifications

ACCURACY	SUBSTRATE	FLATNESS
±0.3 mm + 0.2 mm/m (any tag center to nominal grid, at 20 °C)	Aluminum / LDPE / aluminum composite, 6 mm	≤ 0.5 mm per meter, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

1 Specifications · geometry

FEATURE	SPECIFICATION
Feature position tolerance	±0.3 mm + 0.2 mm/m (any tag center to nominal grid, at 20 °C)
Tag pitch, mean	±0.05 mm
Tag-to-gap ratio	0.3
Edge radius	R6 mm

2 Specifications · physical

FEATURE	SPECIFICATION
Substrate	Aluminum / LDPE / aluminum composite, 6 mm
Weight	6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg)
Pattern	Direct UV print, matte, black on white; optical density ≥ 1.6
Flatness	≤ 0.5 mm per meter, board supported on three points
Mounting	4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size

3 Specifications · environment

FEATURE	SPECIFICATION
Operating temperature	−50 to +80 °C
Thermal expansion	≈ 24 µm/(m·°C)
Storage	−20 to +60 °C, dry
Outdoor lifetime	up to 3 years (UV-stable print)
Cleaning	water or ≤ 70 % isopropyl alcohol, soft cloth

4 Deliverables

INCLUDED		SERVICE	
Measurement	Every board is measured before it ships (dimensional, flatness, visual)	Lead time	Made to order · ships in 2–3 weeks
Files	`.opencv.json`, `.kalibr.yaml`, `.ros2.yaml`, `.dxf`	Options	Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE
Documents	this datasheet, CT-UM-GEN user manual		

5 Compliance & customs

TOPIC	STATEMENT
RoHS	RoHS-compliant materials.
HS code	`9031 80` Not a toy. not for food contact.

6 Revision history

REV	DATE	CHANGE
A	`2026-10`	First issue, planning values

7 Disclaimer

Values are nominal unless stated as measured.